Study Questions for Lectures on Hardy-Weinberg Equilibrium

This study guide provides examples for the types of questions that you might see on the first term from the lectures over Hardy-Weinberg Equilibrium and the effects of violating Hardy-Weinberg assumptions. This is NOT an exhaustive list of all content that might be covered on the exam. Use your notes from lecture and the lecture slides and audio on Canvas to know what all topics have been covered.

1. True or False: If the genotype frequencies of a given population add up to 1, then that population is in Hardy Weinberg Equilibrium.
    a. False
    b. True

2. The table below shows the observed genotype frequencies in a population along with the expected genotype frequencies under Hardy-Weinberg equilibrium. Which of the following can you conclude from this data?
    a. The population follows all Hardy-Weinberg assumptions
    b. Inbreeding is likely occurring
    c. There is probably selection in favor of the heterozygote
    d. Drift is likely high in this population

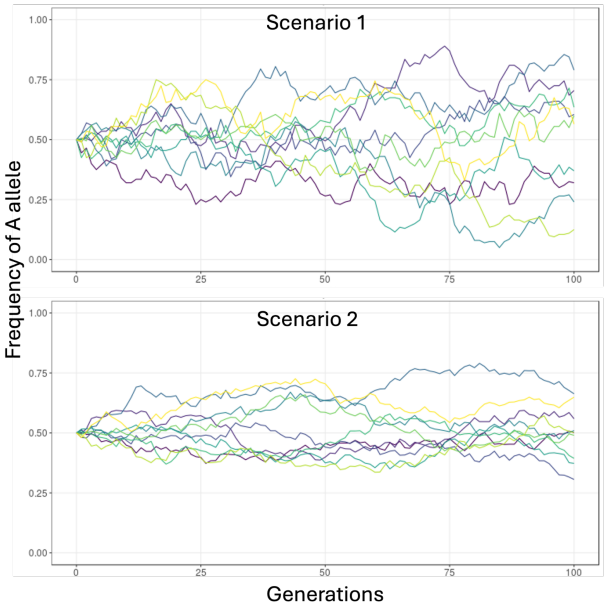
Genotype	Expected	Observed
AA	0.09	0.02
Aa	0.42	0.64
aa	0.49	0.34

3. Which of the following ***is not*** a violation of Hardy-Weinberg assumptions?
    a. Lizards preferentially choose mates with brighter dewlaps
    b. There are ten different alleles for a particular gene
    c. Foxes with tamer personalities are more likely to reproduce
    d. A new allele arises in the population
    e. A group of five bats colonize a new, previously uninhabited island

4. A gene for flower color in pea plants has two alleles, A and a. Given the following genotype frequencies, which of these populations shows evidence of ***possible inbreeding***?
    a. AA=0.01, Aa=0.30, aa=0.69
    b. AA=0.04, Aa=0.32, aa=0.64
    c. AA=0.25, Aa=0.5, aa=0.25
    d. AA=0.20, Aa=0.39, aa=0.41
    e. AA=0.09, Aa=0.42, aa=0.49

5. In many large mammal populations, highways can be barriers to movement - individuals rarely cross them. One solution to this problem has been to install overhead bridges for wildlife to safely cross. The goal is to increase gene flow between the populations on either side of the highway. Which of the following would you expect to see if the bridge is successful in its goal?
    a. Higher levels of drift would occur in both populations after the bridge is installed
    b. The two populations would have different mutation rates
    c. The two populations would become more genetically similar
    d. Both populations would decrease in size
    e. More inbreeding would occur in both populations

6. We conducted computer simulations under two different scenarios, shown in the two figures below. Each line represents a separate population simulated under that scenario. The figure below shows how the frequency of the A_1 allele (y axis) changes over time (x axis). Which of the following can you conclude based on this data (select all that apply)?
    a. Scenario 1 represents a smaller population size than Scenario 2
    b. The phenotype produced by the A allele is beneficial in Scenario 1
    c. A new mutation arose in this population in Scenario 1
    d. The phenotype produced by the A allele is beneficial in Scenario 2



7. You are studying a population of grasshoppers and you find four different alleles at the *A* gene, with frequencies shown in the table. Assuming the population is in Hardy-Weinberg equilibrium, the frequency of genotype A_1A_3 is _____ and the frequency for genotype A_2A_2 is _____.

Allele	Frequency
A_1	p
A_2	q
A_3	r
A_4	s

8. A small population of 26 individuals has five alleles, *A*, *B, C, D,* and *E* for a particular gene. Five individuals are *DA* heterozygotes, five individuals are *AA* homozygotes, five individuals are *AB* heterozygotes, five individuals are *CD* heterozygotes, five individuals are *CC* homozygotes, and one individual is an *EE* homozygote.
    a. What is the frequency of each allele in this population?

    b. If five *AE* heterozygotes migrate into the population, what will be the resulting allele frequencies?

9. A population of otters in H-W equilibrium has two alleles for fur color: *D* is dark brown and complete dominant to *d*, which is light brown. If 81% of the population has light brown fur, what is the frequency of homozygous dominant otters?

10. A population of plants in H-W equilibrium has two alleles (*B* and *b*) for flower color. The frequency of the *B* allele in the population is 0.7. What is the frequency of plants in this population that are heterozygous for flower color?

11. A nearby population of snails in H-W equilibrium has two alleles (*R* and *r*) for shell color. If the frequency of snails that are homozygous for *R* is 0.64, what is the frequency of snails that are homozygous for *r*?

12. A population of trees in H-W equilibrium has two alleles (*F* and *f*) for seed color. The frequency of trees heterozygous for seed color is 0.42. You know that the *f* allele is more common than the *F* allele in this population. What are the frequencies of each allele in this population?

13. A population of geckos in H-W equilibrium has two alleles for tail length: *L* for long tails, which is incomplete dominant to *l* for short tails. The frequency of geckos with long tails is 0.36. What percent of the population would you expect to be heterozygous?

14. A population of bison in H-W equilibrium has three alleles for horn shape: long (*A*), short (*a*) and curly (α). The long allele is dominant over both the short and curly alleles, and the short allele is dominant over the curly allele. The frequency of the *a* allele is 0.5, and 16% of the population has curly horns.
    a. What percent of the population has long horns? *Hint: remember that long horns is a phenotype that represents multiple genotypes*
    b. Which of the three phenotypes (long, short, curly) is the most common in this population?
    c. What is the frequency of homozygotes in this population?

15. Given the following genotype frequencies, is each of these populations in Hardy-Weinberg equilibrium? If not, determine which observed genotype frequencies are greater or less than expected and propose an explanation for the observed phenotypes (ie. Which H-W assumption is violated?). *Note: there may be more than one possible explanation for the observed genotype frequencies*
    a. AA = 0.01, Aa = 0.18, aa = 0.81
    b. AA = 0.25, Aa = 0.50, aa = 0.25
    c. AA = 0.36, Aa = 0.60, aa = 0.04
    d. AA = 0.09, Aa = 0.42, aa = 0.49
    e. AA = 0.64, Aa = 0.20, aa = 0.16

16. How are the founder effect and a population bottleneck similar? How are they different? Which one would you predict to have a more serious negative impact on the species as a whole?

17. Describe the consequences of genetic drift. Why are small populations more affected by drift than larger populations?

18. Gene flow and genetic drift are sometimes referred to as 'non-adaptive evolution.' What does this phrase mean? What type(s) of evolution is/are adaptive?

19. Describe the genetic mechanism by which inbreeding increases homozygosity in a population.

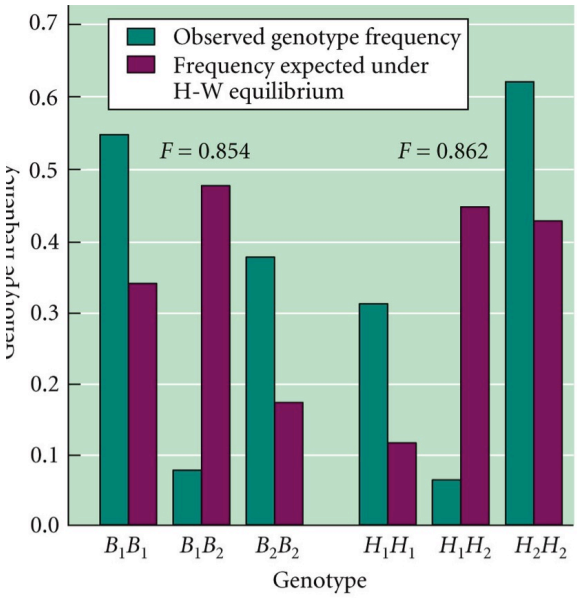
    a. If a population of plants begins with two alleles being equally common (p=0.5 and q=0.5) and all plants self-fertilize, what would you expect the frequency of heterozygotes to be in the first generation? In the 50th? In the 100th? *Hint: there's a formula for this – look for a pattern in the first few generations.*

    b. After how many generations would you predict there will be no heterozygotes in the population?

20. We listed five conditions that a population must meet in order to be in Hardy-Weinberg Equilibrium. How might violation of one assumption lead to or be affected by violations of other assumptions? For example, how might natural selection affect the fate of a new mutation in a population?

21. The figure at right shows the observed and expected genotype frequencies for a population of *Avena fatua* grass.
    a. Is this population in Hardy-Weinberg Equilibrium? If not, describe what assumptions of Hardy-Weinberg Equilibrium could be violated in this population. *Hint: there is more than one possible answer*

    b. Are the two genes experiencing the same pattern? How does your answer to that question narrow down the possible options you listed in part A?



22. The table below shows the genotype frequencies for a particular gene observed in a fictional population of otters, and the expected genotype frequencies if this population was in Hardy-Weinberg Equilibrium. What violations of HWE conditions could be affecting this population? *Select all that apply*

	RR	*Rr*	*rr*
Observed Frequencies	0.43	0.21	0.36
Expected Frequencies under HWE	0.15	0.19	0.66

a. Migration of *rr* individuals out of the population
b. Genetic drift increasing the frequency of the *R* allele
c. Inbreeding
d. Directional selection in favor of the *RR* phenotype
e. Mutation of *R* alleles into *r* alleles